Reg. No. | | | | | | | | | | | [12EC56/13EC56]

FIFTH SEMESTER B.E/B.TECH EXAMINATION, DECEMBER–2015

CONTROL ENGINEERING

Time:3 Hours **Maximum Marks:70**

Instructions:1. Answer any **FIVE** full Questions.

2. Missing data may be suitably assumed.

1. a. Obtain F-I analogous system for the system shown below: **06**

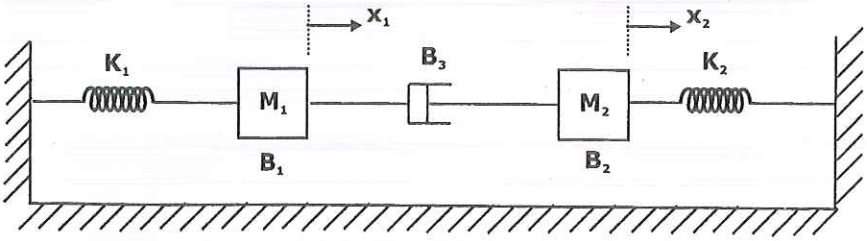

Fig. 1(a)

b. Obtain F-V analogous system for the rotational system shown below. **08**

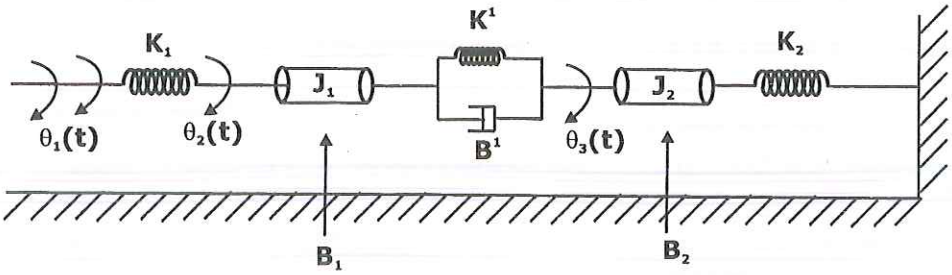

Fig. 1(b)

2. a. Use Masons gain formula to find $\frac{C(s)}{R(s)}$. **08**

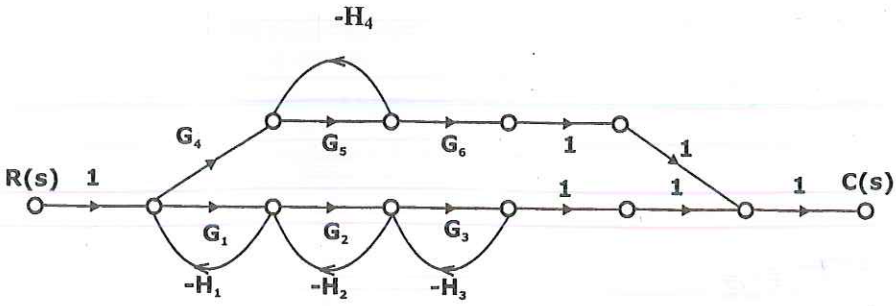

Fig. 2(a)

b. Reduce the block diagram using block diagram reduction rules and obtain overall transfer function. **06**

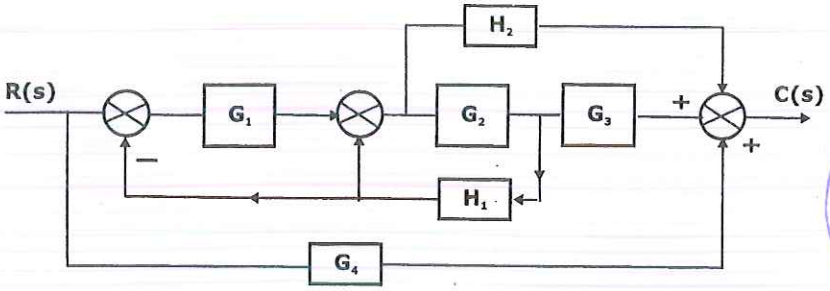

Fig. 2(a)

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3. a. A system is given by differential equation, $\frac{d^2y}{dt^2}+4\frac{dy}{dt}+8y=8x$ Where y=output and x=input. Determine all time domain specifications for unit step input. **08**

b. For a system $G(s)H(s)=\frac{K}{S^2(s+2)(s+3)}$. find the value of K to limit steady state error to 10 when input to the system is $1+10t+\frac{40}{2}t^2$ **06**

4. a. Comment on stability using Routh's criteria $S^5+S^4+2S^3+2S^2+3S+15=0$. **08**

b. List advantages of Routh's criteria. **06**

5. a. For a unity feedback system $G(s)=\frac{K}{S(s+4)(s+2)}$. Sketch the root locus and comment on stability of the system. **14**

6. a. Sketch the Bode plot for the system having $G(s)H(s)=\frac{K(s+0.2)}{s(s+0.01)(s+2)(s+10)}$. Determine the marginal value of K. **14**

7. a. Explain the following terms: **06**
   - i. State variables
   - ii. State transition matrix
   - iii. State transition equation.

b. Obtain the state model for system represented by **08**

$$\frac{d^3y}{dt^3}+6\frac{d^2y}{dt^2}+11\frac{dy}{dt}+10y=3u(t)$$

8. a. Check controllability of the following system **07**

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}-3 & 1\\ -2 & 1.5\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}+\begin{bmatrix}1\\ 4\end{bmatrix}u$$

b. Show that following system is not completely observable **07**

$$\dot{x}=Ax+Bu$$
$$y=Cx$$

Where

$$x=\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}\quad A=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -6 & -11 & -6\end{bmatrix}\quad B=\begin{bmatrix}0\\ 0\\ 1\end{bmatrix}\quad C=\begin{bmatrix}4 & 5 & 1\end{bmatrix}$$